Exercise 2 – Component Communication and Lifecycle Hooks

Objective

The objective of this exercise is to understand Angular component communication, reusable components, lifecycle hooks, conditional rendering, list rendering, and event handling. Students will learn how to pass data between components using @Input() and @Output(), use Angular lifecycle methods, and dynamically render data using structural directives.

Component Communication in Angular

Angular applications are composed of multiple components that communicate with each other.

There are two common communication mechanisms:

Parent → Child Communication

The parent component passes data to a child component using the @Input() decorator.

Example:

```
@Input()

course:any;
```

Child → Parent Communication

The child component sends events back to the parent using the @Output() decorator and an EventEmitter.

Example:

```
@Output()

enrollRequested = new EventEmitter<number>();
```

Angular Lifecycle Hooks

Lifecycle hooks allow developers to execute code at different stages of a component's life.

Lifecycle Hook	Purpose
ngOnInit()	Executes when the component is initialized
ngOnChanges()	Executes when input properties change
ngOnDestroy()	Executes before the component is destroyed

Updating the Home Component

home.component.ts

```
import {
Component,
OnInit,
OnDestroy
} from '@angular/core';
@Component({
selector:'app-home',
standalone:true,
templateUrl:'./home.component.html',
styleUrls:['./home.component.css']
})
export class HomeComponent
implements OnInit,OnDestroy{
portalName="Student Course Portal";
message="Welcome to Angular";
isPortalActive=true;
searchTerm="";
students=350;
courses=25;
```

```
faculty=18;

ngOnInit(){

console.log("Home Component Initialized");

}

ngOnDestroy(){

console.log("Home Component Destroyed");

}

onEnrollClick(){

alert("Successfully Enrolled");

}

}
```

home.component.html

```
<h2>{{portalName}}</h2>

<p>{{message}}</p>

<div *ngIf="isPortalActive">

Portal Status

Active

</div>

<input

[(ngModel)]="searchTerm"

placeholder="Search Courses">

<p>

Searching :

{{searchTerm}}

</p>

<button
```

```
(click)="onEnrollClick()">

Enroll

</button>
```

Course Card Component

The Course Card component is reusable and displays information about a single course.

course-card.component.ts

```
import {

Component,

Input,

Output,

EventEmitter,

OnChanges,

SimpleChanges

} from '@angular/core';


@Component({

selector:'app-course-card',

standalone:true,

templateUrl:'./course-card.component.html',

styleUrls:['./course-card.component.css']

})


export class CourseCardComponent

implements OnChanges{

@Input()
```

```
course:any;
@Output()
enrollRequested=
new EventEmitter<number>();
ngOnChanges(
changes:SimpleChanges
){
console.log(
changes
);
}
enroll(){
this.enrollRequested.emit(
this.course.id
);
}
}
```

---

course-card.component.html

```
<div class="card">
<h3>
{{course.name}}
</h3>
<p>
Code :
{{course.code}}
</p>
<p>
```

```
Credits :
{{course.credits}}
</p>
<button
(click)="enroll()">
Enroll
</button>
</div>
```

---

course-card.component.css

```
.card{
padding:20px;
border:1px solid gray;
border-radius:10px;
margin:15px;
box-shadow:0 2px 4px rgba(0,0,0,.2);
}
```

Updating Course List Component

course-list.component.ts

```
import {
Component
} from '@angular/core';

@Component({
selector:'app-course-list',
standalone:true,
```

```
templateUrl:'./course-list.component.html',

styleUrls:['./course-list.component.css']

})

export class CourseListComponent{

selectedCourseId=0;

courses=[

{

id:1,

name:'Angular',

code:'ANG101',

credits:4

},

{

id:2,

name:'React',

code:'RCT201',

credits:3

},

{

id:3,

name:'Spring Boot',

code:'SPR301',

credits:5

},

{

id:4,

name:'Java',

code:'JAVA201',
```

```
credits:4
},
{
id:5,
name:'MySQL',
code:'SQL101',
credits:2
}
];
onEnroll(id:number){
this.selectedCourseId=id;
}
}
```

---

course-list.component.html

```
<h2>
Available Courses
</h2>
<app-course-card
*ngFor="let course of courses"
[course]="course"
(enrollRequested)="onEnroll($event)">
</app-course-card>
<p>
Selected Course :
{{selectedCourseId}}
</p>
```

Updating Student Profile

student-profile.component.ts

```
import { Component } from '@angular/core';

@Component({

selector:'app-student-profile',

standalone:true,

templateUrl:'./student-profile.component.html'

})

export class StudentProfileComponent{

student={

id:101,

name:'John Doe',

branch:'Computer Science',

semester:5,

cgpa:8.7

};

}
```

---

student-profile.component.html

```
<h2>

Student Profile

</h2>

<table>

<tr>

<td>ID</td>

<td>{{student.id}}</td>

</tr>

<tr>
```

```
<td>Name</td>

<td>{{student.name}}</td>

</tr>

<tr>

<td>Branch</td>

<td>{{student.branch}}</td>

</tr>

<tr>

<td>Semester</td>

<td>{{student.semester}}</td>

</tr>

<tr>

<td>CGPA</td>

<td>{{student.cgpa}}</td

</tr>

</table>
```

Output Screenshot: